SIMATS ENGINEERING
ASSESSMENT TOOL 2 - MEDIUM (Assignment)

COURSE CODE: CSA09 **COURSE NAME: Programming in Java**

COURSE OUTCOME COVERED

CO2: *Implement Collection Framework interfaces such as List, ArrayList, Set, Map, HashTable, and HashMap using iterators and generics to store, retrieve, and process groups of objects in web applications.(BL3,BL4)*

Assessment Tool Weightage: 20%

QUESTIONS: 3 **Total Marks: 30**

ANNEXURE A

Assignment

Code of Conduct:

I Kandath Haneesh (192324084) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

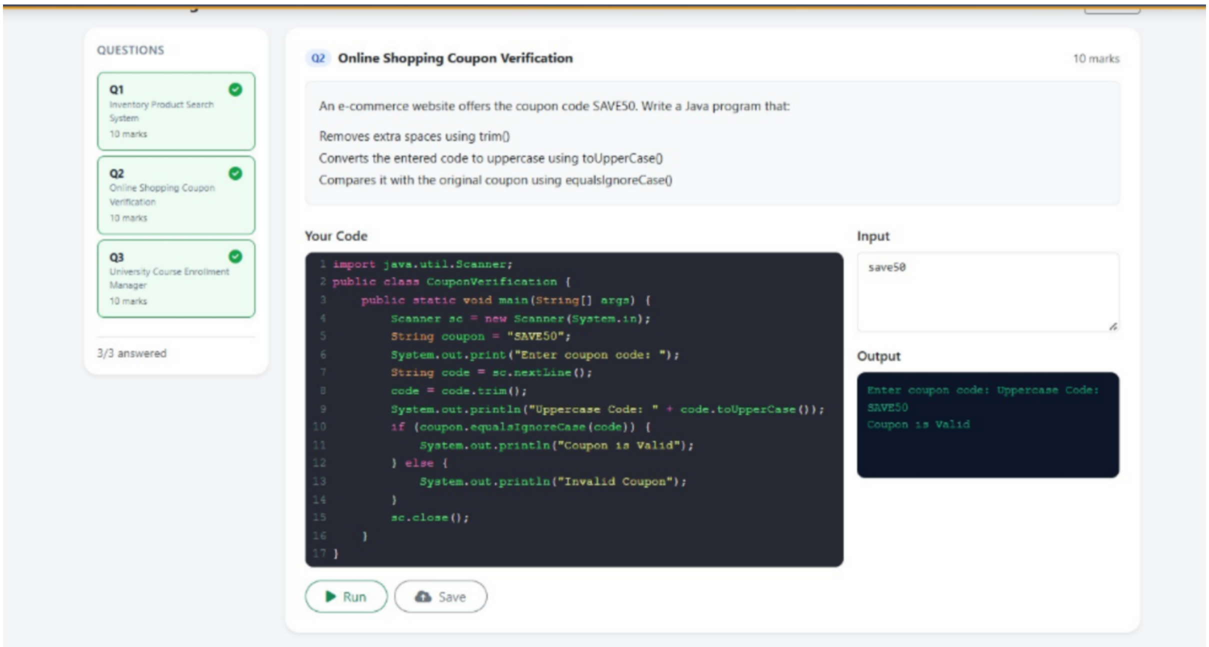
Signature of the student:Kandath Haneesh

ANNEXURE

Assignment

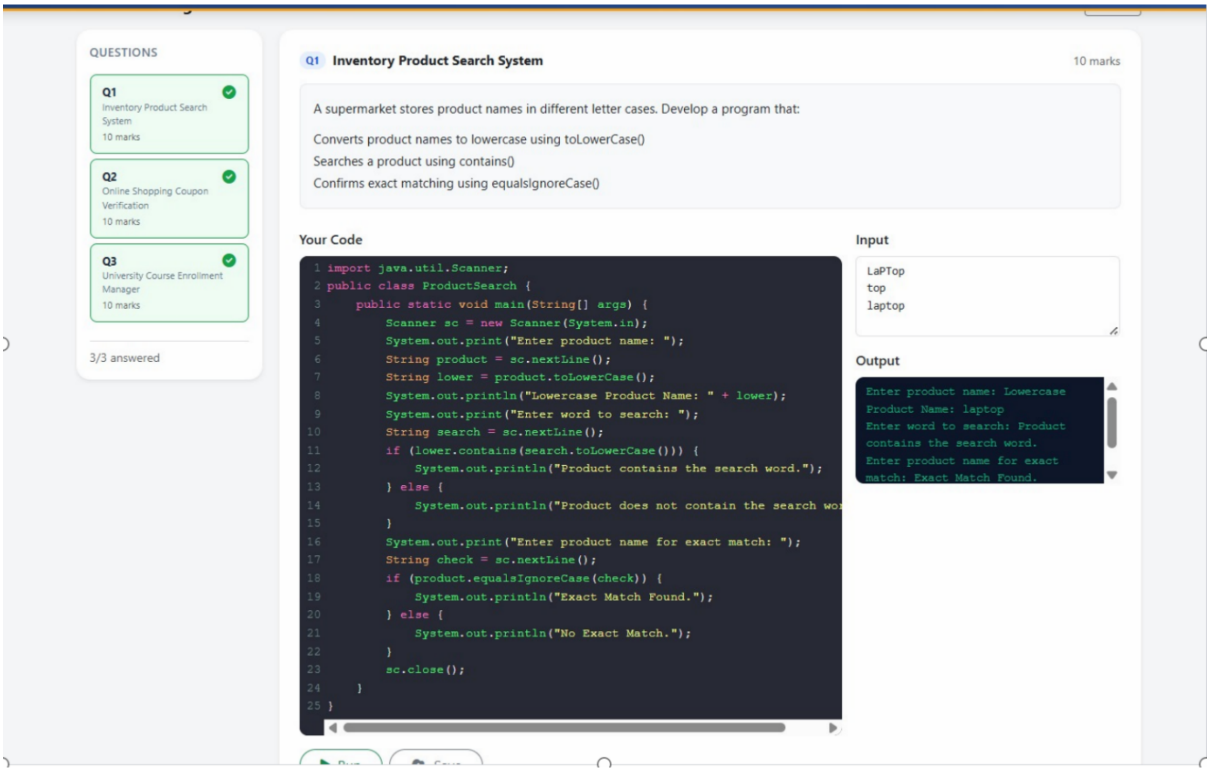
1. **Online Shopping Coupon Verification**
   "An e-commerce website offers the coupon code SAVE50. Write a Java program that:

Removes extra spaces using trim()
Converts the entered code to uppercase using toUpperCase()
Compares it with the original coupon using equalsIgnoreCase()"



2. **Inventory Product Search System**
"A supermarket stores product names in different letter cases. Develop a program that:

Converts product names to lowercase using toLowerCase()
Searches a product using contains()
Confirms exact matching using equalsIgnoreCase()"



3. University Course Enrollment Manager

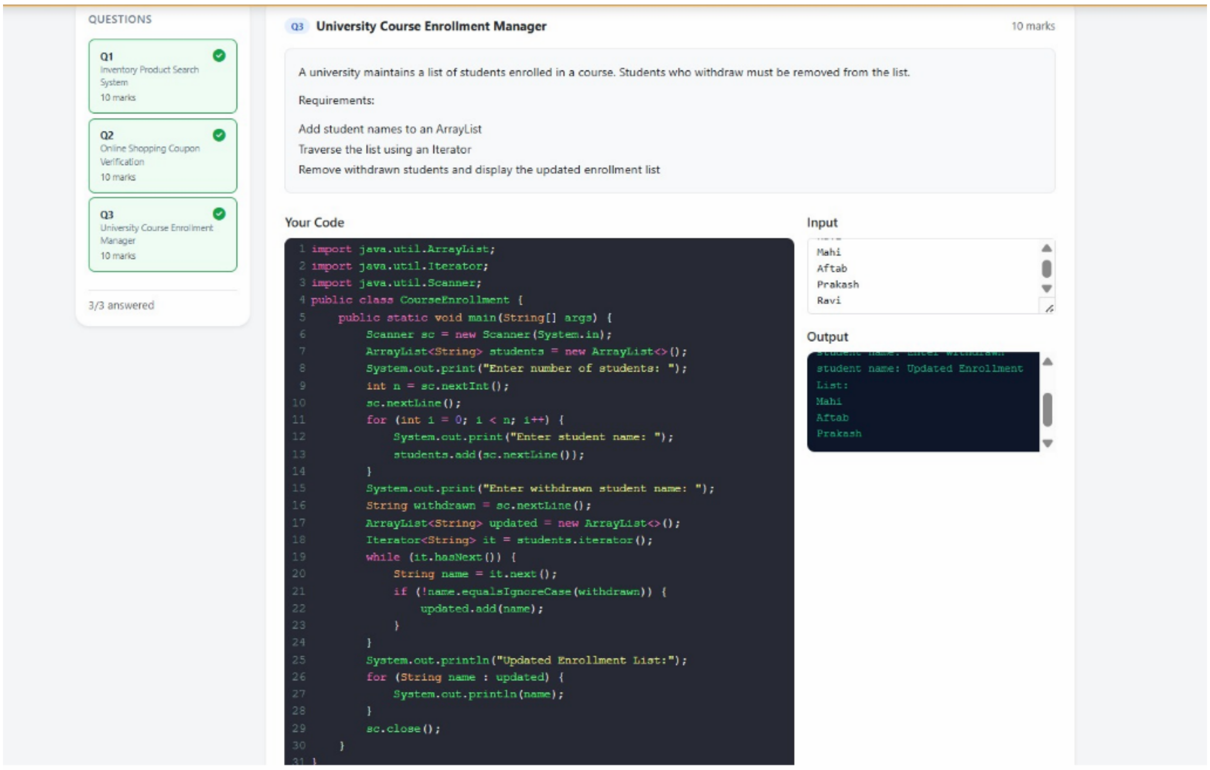
"A university maintains a list of students enrolled in a course. Students who withdraw must be removed from the list.

Requirements:

Add student names to an ArrayList

Traverse the list using an Iterator

Remove withdrawn students and display the updated enrollment list"



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30	Applies appropriate concepts effectively to solve complex real-world problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & Decision-Making Skills	20	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification